The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 April 1993

INCH-POUND

MIL-S-19500/500B
19 January 1993
SUPERSEDING
MIL-S-19500/500A
14 September 1981

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, TRANSIENT VOLTAGE SUPPRESSOR
TYPES 1N5555 THROUGH 1N5558, 1N5907, 1N5629A THROUGH 1N5665A
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for 1500 watt, peak, pulse power, silicon, transient, voltage suppressor diodes. Four levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. Maximum ratings are as shown in columns 5 through 8 of table III herein, and as follows:

$P_{PP}$ = 1500 W (see figure 3) at $t_p$ = 1.0 ms.

$P_{M(AV)}$ = 1.0 W (derate at 6.67 mW/°C above $T_A$ = +25°C) (see 6.3).

$I_{FSM}$ = 200 A (pk) at $t_p$ = 8.3 ms ($T_A$ = +25°C).

-55°C ≤ $T_J$ ≤ +175°C (ambient), -55°C ≤ $T_{STG}$ ≤ +175°C (ambient).

1.4 Primary electrical characteristics at $T_A$ = +25°C. Primary electrical characteristics are shown in columns 2 and 4 of table III herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPD), NASA Goddard Space Flight Center, Code 310.A, Greenbelt, MD 20771 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 5961

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

$V_C$ - - - - - - - - - - - - Clamping voltage. The maximum peak voltage appearing across the device when subjected to the peak pulse current $I_{PP}$.

$I_{PP}$ - - - - - - - - - - - - Peak pulse current at a specified condition.

$I_{(BR)}$ - - - - - - - - - - - - Reverse breakdown current at a specified condition.

$P_{PP}$ - - - - - - - - - - - - Repetitive peak pulse power.

$\alpha V_{(BR)}$ - - - - - - - - - - - - Temperature coefficient of $V_{(BR)}$.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and figure 1 herein.

3.3.1 Metallurgical bond construction. Metallurgically bonded construction is required. The bonding metal shall have flow points above 260°C.

3.3.2 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein except, lot accumulation period shall be six months in lieu of six weeks.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.



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4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement | |
|---|---|---|
| | JANS level | JANTX and JANTXV levels |
| 3 | $T_{(high)}$ = +175°C | $T_{(high)}$ = +175°C |
| 9 | Not applicable | Not applicable |
| 10 | Not applicable | Not applicable |
| 11 | Not applicable | Not applicable |
| 12 | See 4.5.1 | See 4.5.1 |
| 13 | Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12). Subgroup 3 of table I herein. | Interim electrical, delta, and group A, subgroup 2, electrical parameters not applicable for this screen (performed in screen 12). |

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. End-point electrical measurements shall be in accordance with the applicable steps of table II herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

a. Subgroup 3: Condition for surge, 1 cycle, see 4.5.3b.

b. Subgroup 5: Condition for accelerated steady-state operation life are as follows: See 4.5.2, $T_A$ = +100°C (min); $T_J$ = +225°C (min); t = 168 hours minimum.

c. Subgroup 4 and 6: Not applicable.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX and JANTXV) of MIL-S-19500.

a. Subgroup 2: Condition for surge, 1 cycle, see 4.5.3b.

b. Subgroup 3: Condition for steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3b).

c. Subgroup 5: Not applicable.



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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

a. Subgroup 2: Condition for terminal strength are as follows: Lead tension: Test condition A; weight = 5 lbs; t = 15 ±3 s. Lead fatigue: Test condition E; weight = 8 oz.

b. Subgroup 6: Condition for steady-state operation life are as follows: See 4.5.1, 1 ms pulse only (see 4.5.3b).

c. Subgroup 7: Condition for temperature coefficient of breakdown voltage are as follows: $I_{(BR)}$ = column 3 of table III, $T_1$ = +25°C ±3°C, $T_2$ = $T_1$ +100°C, LTPD = 10. (All electrical measurements and test shall be performed twice, once in each direction.)

d. Subgroup 8: Condition for maximum peak pulse current are as follow: See 4.5.3a, (20 μs pulse only) 10 pulses, LTPD 10.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables as follows.

4.5.1 Power burn-in (HTRB) and steady-state operation life test conditions. The test conditions and order of events shall be as follows:

a. Read and record $I_D$ and $V_{(BR)}$ at $T_A$ = +25°C.

b. Pulse in accordance with 4.5.3b, 20 times (screening and Group B) and 100 times (group C) at $T_A$ = +25°C.

c. Read $I_D$ at $T_A$ = +25°C, remove defective devices and record the number of failures.

d. Apply the working peak reverse voltage ($V_{WM}$) (column 4 of table III) at an ambient temperature of +125°C as follows:

1. 96 hours (JANTX and JANTXV) and 240 hours (JANS) for the screening test.

2. 340 hours (JAN, JANTX, and JANTXV) for group B steady-state operation life test.

3. 1,000 hours for group C steady-state operation life test.

e. Read and record $I_D$ and $V_{(BR)}$ at $T_A$ = +25°C. Devices with $\Delta I_D$ > 50 percent (100 percent for steady-state operation life) of the initial reading or 1 μA dc, whichever is greater, or $\Delta V_{(BR)}$ > ±2 percent (±5 percent for steady-state operation life) initial value shall be considered defective. Remove defective devices and record the number of failures. (For steady-state operation life, $I_D$ maximum limit shall be two times the group A limit).

4.5.1.1 Group C steady-state operation life test (alternate procedure). When the group B 340 hour life test is continued on test to 1000 hours to satisfy the group C life test requirements, the test shall be performed as given in 4.5.1 with the following exceptions:

a. 4.5.1 step b shall be moved and performed following step e.

b. 4.5.1 step e shall be repeated after step b is performed.



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4.5.2 Accelerated steady-state operation life (JANS). This test shall be conducted with the devices subjected to the breakdown current specified in column 10 of table III for a total duration of 168 hours. At the beginning and end (with devices cooled to $T_A$ = +25°C) of the test, the devices shall be subjected to pulse conditions at the rate of one pulse per minute (max) for 10 pulses in accordance with 4.5.3 as specified.

4.5.3 Maximum peak pulse current ($I_{pp}$). The peak pulse currents specified in column 7 of table III shall be applied in the reverse direction while simultaneously maintaining a reverse bias voltage of not less than the applicable voltage specified in column 4 of table III. The peak current shall be applied with a current VS time waveform as follows (1 pulse per minute maximum):

a. Pulse current shall reach 100 percent of $I_{pp}$ at $t \leq 8\ \mu s$ and decay to 50 percent of $I_{pp}$ at $t \geq 20\ \mu s$ for $t_p = 20\ \mu s$, (see figure 5).

b. Pulse current shall reach 100 percent of $I_{pp}$ at $t \leq 10\ \mu s$ and decay to 50 percent of $I_{pp}$ at $t \geq 1$ ms for $t_p = 1$ ms (see figure 4).

NOTE: Tolerance on time (t) shall be -0, +10 percent.

4.5.4 Clamping voltage. The peak pulse clamping voltage shall be measured across the diode in a 1 ms time interval. The response detector shall demonstrate equipment accuracy of ±3 percent. The peak clamping voltage as specified in column 6 of table III shall be applicable to the 1 ms pulse of paragraph 4.5.3b only.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as specified (see 3.3.2).

c. Type designation and product assurance level.

6.3 Steady-state power rating. This rating is not relevant for most applications.

6.4 Substitution information. Type numbers 1N5555 through 1N5558 were previously covered by MIL-S-19500/434.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits 2/ | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical examination | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Reverse current | 4016 | DC method, $V_R = V_{WM}$ (column 4 of table III) | $I_R$ | | column 5 | µA dc |
| Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle $\leq 2$ percent, $I_{(BR)}$ = column 3 of table III | $V_{(BR)}$ | column 2 | | V dc |
| Subgroup 3 | | | | | | |
| Minimum breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle $\leq 2$ percent, $I_{(BR)}$ = column 3 of table III, $T_A = -55°C$ | $V_{(BR)}$ | column 9 | | V dc |
| Subgroup 4 | | | | | | |
| Clamping voltage maximum (pulsed) (see 4.5.4) | | $t_p = 1.0$ ms (see 4.5.3b), $I_{PP}$ = column 7 of table III | $V_C$ | | column 6 | V (pk) |
| Forward voltage | 4011 | $I_{FM} = 100$ A (pk), $t_p = 8.3$ ms duty (max), cycle = 4 pulses per minute (max) | $V_{FM}$ | | 3.5 | V (pk) |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |
| Subgroup 6 | | | | | | |
| Forward surge current | 4066 | $I_{FSM} = 200$ A (pk), one pulse, half sinewave (8.3 ms), $I_F = 0$; $V_{WM} = 0$; $T_A = 25°C$ | | | | |
| End-point electrical measurements | | See table II, steps 1 and 2 | | | | |

1/ For sampling plan, see MIL-S-19500.

2/ Column references are to table III.



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TABLE II. Groups B and C electrical measurements. 1/ 2/ 3/

| Step | Inspection | MIL-STD-750 | | Symbol | Limits 4/ | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1 | Reverse current | 4016 | DC method, $V_R = V_{WM}$ column 4 of table III | $I_R$ | | column 5 | µA dc |
| 2 | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III | $V_{(BR)}$ | column 2 | | V dc |
| 3 | Forward voltage | 4011 | $I_{FM}$ = 100 A (pk), $t_p$ = 8.3 ms (max) at a pulse rate of one pulse per minute (max) | $V_{FM}$ | | 3.5 | V (pk) |
| 4 | Standby current | 4016 | DC method, $V_R = V_{WM}$ (column 4 of table III | $\Delta I_D$ 5/ | | 100 percent of initial reading or 20 percent of column 5 of table III, whichever is greater. | |
| 5 | Breakdown voltage | 4022 | $t_p \leq 300$ ms, duty cycle ≤ 2 percent, $I_{(BR)}$ = column 3 of table III | $\Delta V_{(BR)}$ 2/ | | ±5 percent of initial value | |
| 6 | Clamping voltage maximum (pulsed) | | $t_p$ = 1.0 ms, (see 4.5.3b), $I_{PP}$ = column 7 of table III | $V_C$ | | column 6 | V (pk) |

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1, 2, 3, and 6.

b. Subgroup 5, see table II herein, steps 1, 2, 3, 4, and 5.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2.

b. Subgroup 6 see table II herein, steps 1, 2, 4, and 5.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2 and 3, see table II herein, steps 1, 2, and 3 for JANS and steps 1 and 2 for JANTX and JANTXV.

b. Subgroup 8, see table II herein, steps 1 and 2 for all levels.

4/ Column references are to table III.

5/ Devices which exceed the group A limits for this test shall not be accepted.



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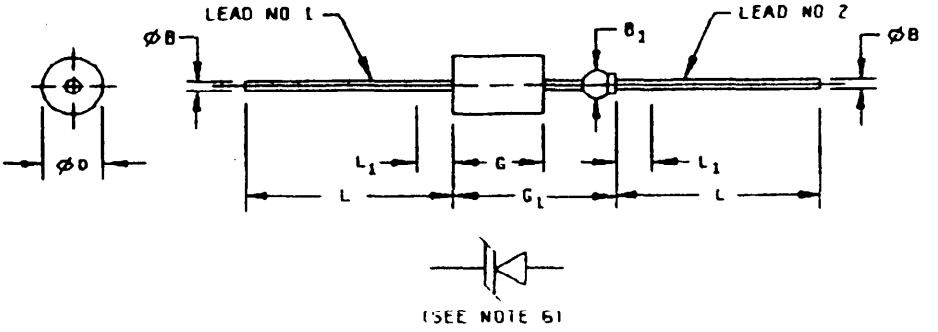
TABLE III. Characteristics and Ratings.

| Col 1 | Col 2 | | Col 3 | Col 4 | Col 5 | Col 6 | Col 7 | | Col 8 | Col 9 | Col 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Type | Breakdown voltage $V_{(BR)}$ at $I_{(BR)}$ | | Test current $I_{(BR)}$ | Working peak reverse voltage $V_{WM}$ | Maximum reverse current $I_R$ | Maximum clamping voltage $V_C$ at $I_{PP}$ for $t_p$ = 1 ms | Maximum peak pulse current ($I_{PP}$) | | Maximum temperature coefficient of $V_{(BR)}$ $\alpha V_{(BR)}$ | Minimum breakdown voltage at $I_{(BR)}$ $T_A$ = -55°C | Breakdown current ($I_{(BR)}$) maximum dc current $T_A$ = +25°C |
| | Min | Max | | | | | $t_p$ = 20 µs $t_r$ = 8 µs | $t_p$ = 1 ms $t_r$ = 10 µs | | | |
| | V dc | V dc | mA dc | V (pk) | µA dc | V (pk) | A (pk) | A (pk) | α/°C | V dc | mA dc |
| 1N5555 | 33.00 | ---- | 1 | 30.50 | 5 | 47.5 | 193 | 32.0 | .093 | 30.20 | 27.0 |
| 1N5556 | 43.70 | ---- | 1 | 40.30 | 5 | 63.5 | 136 | 24.0 | .094 | 40.00 | 21.0 |
| 1N5557 | 54.00 | ---- | 1 | 49.00 | 5 | 78.5 | 116 | 19.0 | .096 | 48.50 | 16.0 |
| 1N5558 | 191.00 | ---- | 1 | 175.00 | 5 | 265.0 | 33 | 5.7 | .100 | 172.00 | 4.5 |
| 1N5907 | 6.00 | 6.75 | 1 | 5.00 | 300 | 8.5 | 1000 | 150.0 | .057 | 5.63 | 140.0 |
| 1N5629A | 6.45 | 7.14 | 10 | 5.80 | 1000 | 10.5 | 810 | 143.0 | .057 | 6.05 | 140.0 |
| 1N5630A | 7.13 | 7.88 | 10 | 6.40 | 500 | 11.3 | 750 | 132.0 | .061 | 6.66 | 125.0 |
| 1N5631A | 7.79 | 8.61 | 10 | 7.02 | 200 | 12.1 | 700 | 124.0 | .065 | 7.24 | 115.0 |
| 1N5632A | 8.65 | 9.55 | 1 | 7.78 | 50 | 13.4 | 630 | 112.0 | .068 | 8.01 | 104.0 |
| 1N5633A | 9.50 | 10.50 | 1 | 8.55 | 10 | 14.5 | 585 | 103.5 | .073 | 8.75 | 95.0 |
| 1N5634A | 10.50 | 11.60 | 1 | 9.40 | 5 | 15.6 | 545 | 96.0 | .075 | 9.65 | 86.0 |
| 1N5635A | 11.40 | 12.60 | 1 | 10.20 | 5 | 16.7 | 510 | 90.0 | .078 | 10.40 | 73.0 |
| 1N5636A | 12.40 | 13.70 | 1 | 11.10 | 5 | 18.2 | 465 | 82.0 | .081 | 11.30 | 70.0 |
| 1N5637A | 14.30 | 15.80 | 1 | 12.80 | 5 | 21.2 | 400 | 71.0 | .084 | 13.00 | 63.0 |
| 1N5638A | 15.20 | 16.80 | 1 | 13.60 | 5 | 22.5 | 375 | 67.0 | .086 | 13.70 | 59.0 |
| 1N5639A | 17.10 | 18.90 | 1 | 15.30 | 5 | 25.2 | 335 | 59.5 | .088 | 15.40 | 53.0 |
| 1N5640A | 19.00 | 21.00 | 1 | 17.10 | 5 | 27.7 | 305 | 54.0 | .090 | 17.10 | 47.0 |
| 1N5641A | 20.90 | 23.10 | 1 | 18.80 | 5 | 30.6 | 275 | 49.0 | .092 | 18.80 | 43.0 |
| 1N5642A | 22.80 | 25.20 | 1 | 20.50 | 5 | 33.2 | 255 | 45.0 | .094 | 20.50 | 39.0 |
| 1N5643A | 25.70 | 28.40 | 1 | 23.10 | 5 | 37.5 | 225 | 40.0 | .096 | 23.00 | 35.0 |
| 1N5644A | 28.50 | 31.50 | 1 | 25.60 | 5 | 41.4 | 205 | 36.0 | .097 | 23.50 | 31.0 |
| 1N5645A | 31.40 | 34.70 | 1 | 28.20 | 5 | 45.7 | 185 | 33.0 | .098 | 28.00 | 28.0 |
| 1N5646A | 34.20 | 37.80 | 1 | 30.80 | 5 | 49.9 | 170 | 30.0 | .099 | 30.50 | 26.0 |
| 1N5647A | 37.10 | 41.00 | 1 | 33.30 | 5 | 53.9 | 155 | 28.0 | .100 | 33.10 | 24.0 |
| 1N5648A | 40.90 | 45.20 | 1 | 36.80 | 5 | 59.3 | 145 | 25.3 | .101 | 36.40 | 22.0 |
| 1N5649A | 44.70 | 49.40 | 1 | 40.20 | 5 | 64.8 | 130 | 23.2 | .101 | 39.80 | 20.0 |
| 1N5650A | 48.50 | 53.60 | 1 | 43.60 | 5 | 70.1 | 120 | 21.4 | .102 | 43.10 | 18.0 |
| 1N5651A | 53.20 | 58.80 | 1 | 47.80 | 5 | 77.0 | 110 | 19.5 | .103 | 47.30 | 17.0 |
| 1N5652A | 58.90 | 65.10 | 1 | 53.00 | 5 | 85.0 | 100 | 17.7 | .104 | 52.30 | 15.0 |
| 1N5653A | 64.60 | 71.40 | 1 | 58.10 | 5 | 92.0 | 90 | 16.3 | .104 | 57.30 | 14.0 |
| 1N5654A | 71.30 | 78.80 | 1 | 64.10 | 5 | 103.0 | 82 | 14.6 | .105 | 63.20 | 12.0 |
| 1N5655A | 77.90 | 86.10 | 1 | 70.10 | 5 | 113.0 | 75 | 13.3 | .105 | 69.00 | 11.0 |
| 1N5656A | 86.50 | 95.50 | 1 | 77.80 | 5 | 125.0 | 68 | 12.0 | .106 | 76.50 | 10.0 |
| 1N5657A | 95.00 | 105.00 | 1 | 85.50 | 5 | 137.0 | 62 | 11.0 | .106 | 84.10 | 9.5 |
| 1N5658A | 105.00 | 116.00 | 1 | 94.00 | 5 | 152.0 | 55 | 9.9 | .107 | 92.80 | 8.5 |
| 1N5659A | 114.00 | 126.00 | 1 | 102.00 | 5 | 165.0 | 50 | 9.1 | .107 | 100.00 | 7.5 |
| 1N5660A | 124.00 | 137.00 | 1 | 111.00 | 5 | 179.0 | 47 | 8.4 | .107 | 109.00 | 7.0 |
| 1N5661A | 143.00 | 158.00 | 1 | 128.00 | 5 | 207.0 | 40 | 7.2 | .108 | 126.00 | 6.0 |
| 1N5662A | 152.00 | 168.00 | 1 | 136.00 | 5 | 219.0 | 38 | 6.8 | .108 | 134.00 | 5.8 |
| 1N | 162.00 | 179.00 | 1 | 145.00 | 5 | 234 | 36 | 6.4 | .108 | 143.00 | 5.5 |
| 1N. | 171.00 | 189.00 | 1 | 154.00 | 5 | 246 | 34 | 6.1 | .108 | 151.00 | 5.0 |
| 1N5665A | 190.00 | 210.00 | 1 | 171.00 | 5 | 274.0 | 30 | 5.5 | .108 | 167.00 | 4.5 |





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| Symbol | Dimensions | | | | Notes |
|---|---|---|---|---|---|
| | Inches | | Millimeters | | |
| | Min | Max | Min | Max | |
| ØB | .025 | .035 | .64 | .89 | |
| $B_1$ | .045 | .100 | 1.14 | 2.54 | 3 |
| ØD | .215 | .235 | 5.46 | 5.97 | 1 |
| G | .293 | .357 | 7.44 | 9.07 | |
| $G_1$ | ---- | .570 | ---- | 14,48 | |
| L | 1.000 | 1.625 | 25.40 | 41.28 | |
| $L_1$ | ---- | .188 | ---- | 4.78 | 2 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The major diameter is essentially constant along its length.
4. Within this zone, diameter may vary to allow for lead finishes and irregularities.
5. Dimension to allow for pinch or seal deformation anywhere along tubulation.
6. Cathode lead shall be electrically connected to the case. If tubulation is used, it shall be on the anode end.
7. Symbol for internal construction of unipolar transient suppressor.

FIGURE 1. Physical dimensions, type 1N5555 through 1N5558, 1N5907, 1N5629A through 1N5665A (DO-13).



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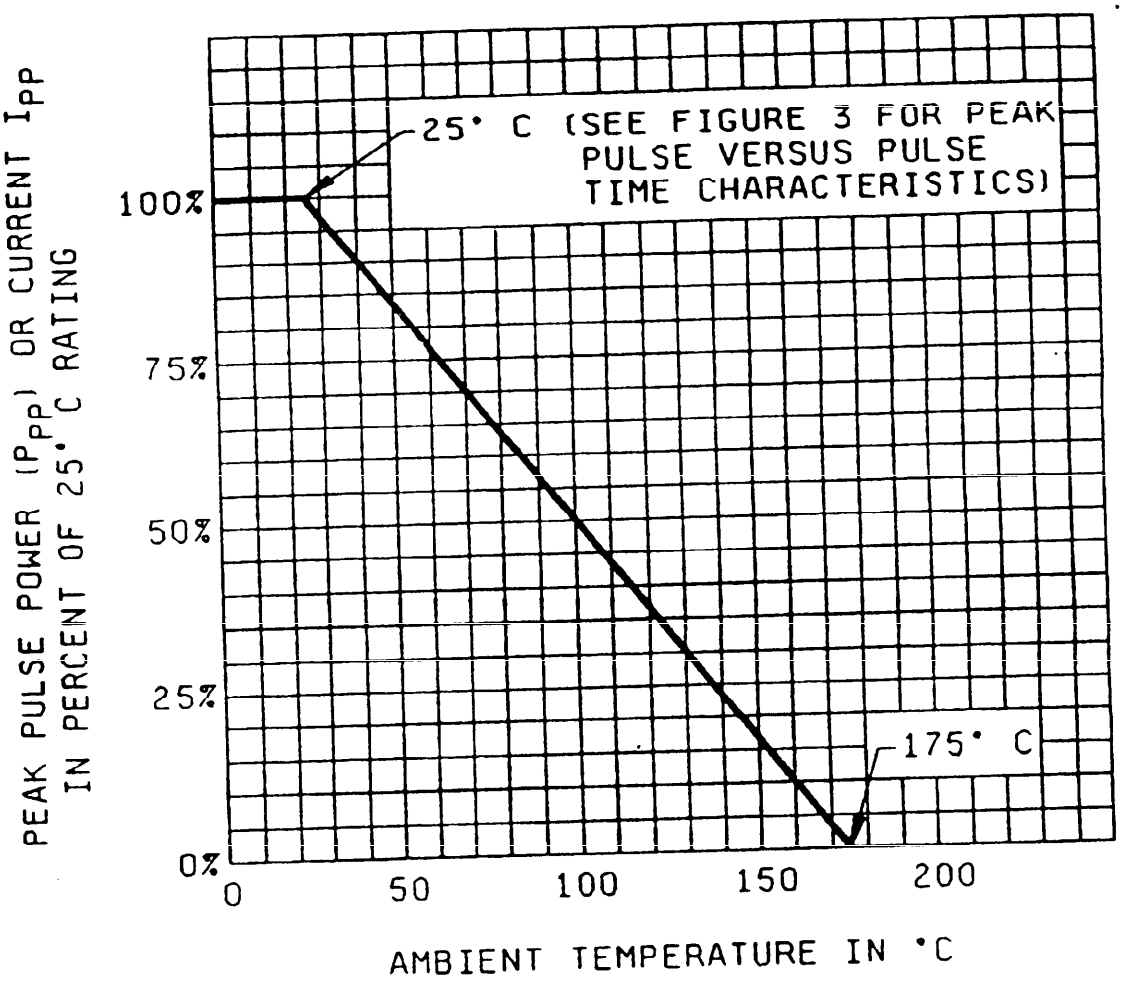



FIGURE 2. Derating curve.



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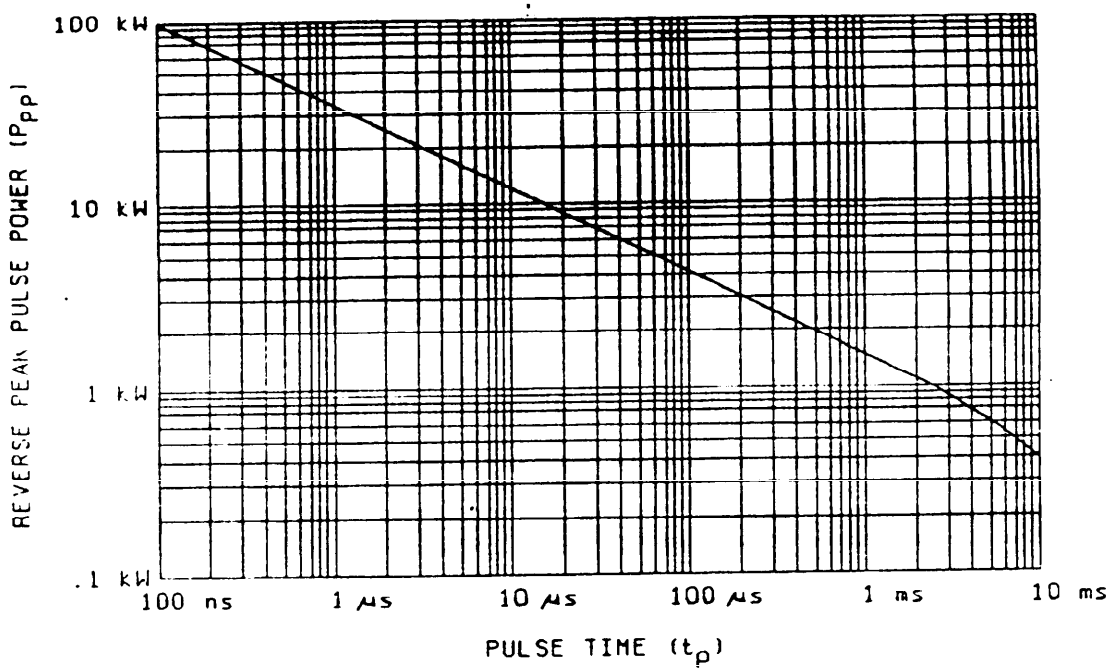
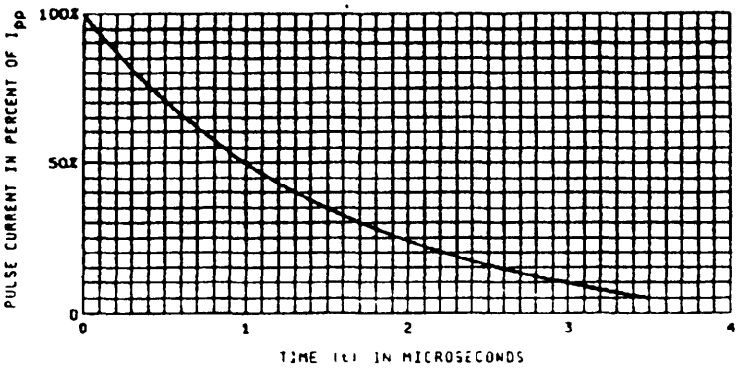



FIGURE 3. Reverse peak pulse power vs. pulse time.

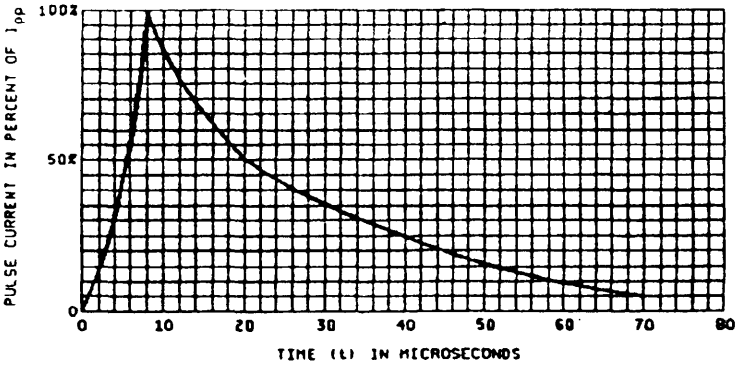


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NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{pp}$. (Rise time to 100 percent of $I_{pp}$ = 10 $\mu$s)

FIGURE 4. Current impulse waveform.




NOTE: Pulse time duration is defined as that point where the pulse current decays to 50 percent of $I_{pp}$. (Rise time to 100 percent of $I_{pp}$ = 8 $\mu$s)

FIGURE 5. Current impulse waveform.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 19, 85
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, SH

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1472)



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